

Internship Report
On
AWS Data Engineering Virtual Internship

Submitted
To
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Assistant Professor

In partial fulfilment of the requirement of degree of
B.Tech CSE

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.



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CANDIDATE'S DECLARATION

I, Namita Sharma do hereby declare that the internship report titled “**AWS Data Engineering Virtual Internship**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature
Namita Sharma
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1. Internship Completion Certificate

This section presents the Internship Completion Certificate awarded upon successful completion of the AWS Data Engineering Virtual Internship conducted through EduSkills Foundation.



Fig 1- Internship Completion Certificate

The above certificate serves as official proof of successful completion of the AWS Data Engineering Virtual Internship through EduSkills Foundation. It confirms the completion of the prescribed internship training and learning activities.

2. Project Description

The internship was a virtual, course-based internship in **AWS Data Engineering** conducted through **EduSkills Foundation**. The main purpose of the internship was to provide knowledge and practical exposure to cloud computing and data engineering concepts using Amazon Web Services (AWS).

During the internship, various concepts related to data storage, data processing, data management, ETL, data pipelines, cloud computing, and data analytics were studied. The training helped in understanding how cloud technologies can be used to handle and process large amounts of data efficiently.

The internship was primarily focused on learning and training rather than developing an independent project. It provided an opportunity to understand AWS services, their applications, and their role in modern data engineering workflows.

The internship helped in connecting academic concepts with industry-oriented technologies and provided a foundation for further learning in **AWS, Cloud Computing, Data Engineering, and Data Analytics**.

2.1 Introduction

Data plays an important role in modern organizations as large amounts of information are generated from applications, websites, transactions, sensors, and other sources. Managing and processing this data efficiently is one of the major challenges faced by organizations.

Data Engineering involves collecting, storing, processing, transforming, and preparing data for analysis. Cloud computing makes these processes more scalable and flexible by providing computing and storage resources over the internet.

Amazon Web Services (AWS) is a leading cloud platform that provides various services for computing, storage, databases, data processing, security, and analytics.

The AWS Data Engineering internship provided an opportunity to understand the fundamentals of cloud-based data engineering and the use of AWS services in managing data. The course-based training included learning materials, practical activities, and assessments that helped develop an understanding of modern data engineering concepts.

2.2 Organization Profile

2.2.1 EduSkills Foundation

EduSkills Foundation is an organization that focuses on providing industry-oriented learning and skill development opportunities to students. It provides various training programs and virtual internship opportunities in emerging technologies.

EduSkills works towards reducing the gap between academic education and industry requirements by providing students with exposure to modern technologies and practical learning experiences.

The organization provides learning opportunities in areas such as **Cloud Computing, Artificial Intelligence, Machine Learning, Data Science, Data Engineering, Cybersecurity, and other emerging technologies.**

The AWS Data Engineering internship was conducted through this learning platform. It provided structured training related to AWS and data engineering concepts and helped students gain knowledge of technologies used in cloud-based environments.

2.3 Problem Statement

Modern organizations generate huge amounts of data from different sources. Managing, storing, processing, and analyzing such data efficiently can be challenging using traditional systems.

Data may also exist in different formats and may require cleaning and transformation before it can be used for analysis. Therefore, organizations require scalable and reliable data engineering solutions that can handle large amounts of data.

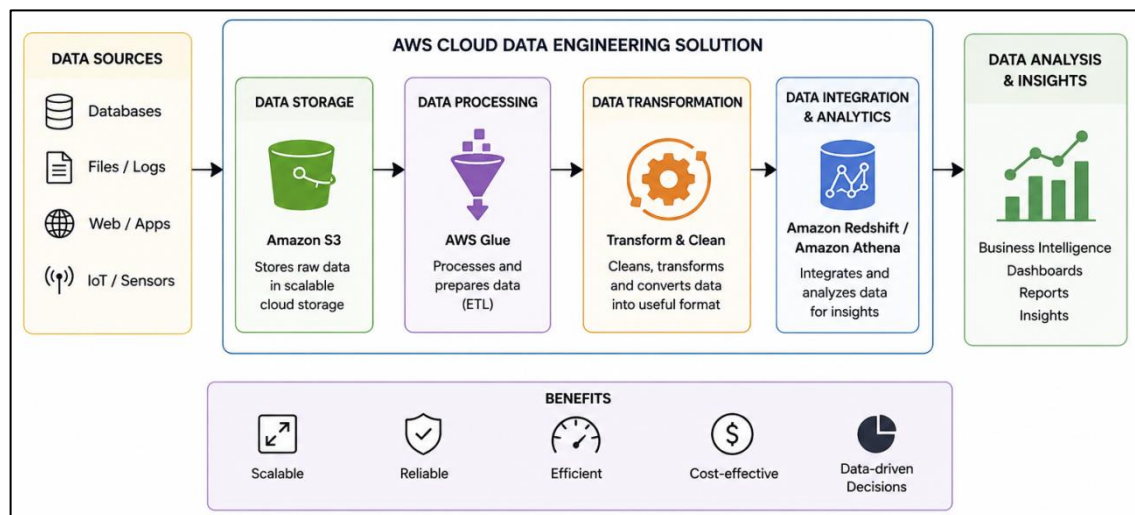


Fig 2 - Diagram Showcasing Problem Statement

The internship focused on understanding how **AWS cloud services can be used for data storage, processing, transformation, integration, and analytics.** The training aimed to provide knowledge of cloud-based data engineering workflows and the technologies used to manage data efficiently.

2.4 Project Objectives

The main objectives of the AWS Data Engineering internship were:

1. To understand the fundamentals of **cloud computing and AWS**.
2. To understand the basic concepts of **Data Engineering**.
3. To learn about cloud-based data storage and processing.
4. To understand **ETL (Extract, Transform, Load)** processes.
5. To understand the concept of data pipelines and data integration.
6. To gain knowledge about AWS services used in data engineering.
7. To understand how processed data can be used for analytics.
8. To gain industry-oriented technical knowledge and practical exposure.
9. To connect academic concepts with real-world cloud technologies.
10. To develop a foundation for further learning in AWS and Data Engineering.

2.5 Scope of The Project

The scope of the internship covers the fundamental concepts of **AWS and Data Engineering**. It includes understanding how data can be collected from different sources, stored in cloud environments, processed, transformed, and prepared for analysis.

The training also covers the basic concepts of **data pipelines, ETL, data storage, data processing, databases, data analytics, and cloud security**.

The knowledge gained during the internship can be useful in various industries such as banking, e-commerce, healthcare, education, telecommunications, and finance, where large amounts of data need to be managed and analyzed.

Overall, the internship provides a foundation for understanding cloud-based data engineering and can be further extended to develop real-world AWS data engineering projects.

2.6 Technologies and Tools Used

During the AWS Data Engineering internship, various cloud and data-related technologies were studied to understand the process of storing, processing, and managing data in a cloud environment.

The major technologies and tools covered during the training include:

- **Amazon Web Services (AWS):** Cloud platform used for learning different cloud-based data engineering services.
- **Amazon S3:** Used for understanding cloud-based object storage and management of data.
- **AWS Glue:** Used to understand data integration, ETL processes, and data transformation.
- **Amazon Redshift:** Used for understanding cloud-based data warehousing and analytical workloads.
- **Amazon Athena:** Used for understanding querying and analyzing data stored in Amazon S3.
- **SQL:** Used for working with structured data and performing data-related queries.
- **Python:** Used for understanding programming and data processing concepts.
- **AWS IAM:** Used to understand identity, access management, and security in AWS.

These technologies helped in developing an understanding of how different AWS services can work together to create a cloud-based data engineering environment.

2.7 System Architecture

The AWS Data Engineering architecture represents the flow of data from its source to storage, processing, and analysis. In a typical cloud-based data engineering workflow, data is first collected from different sources and stored in a cloud storage service.

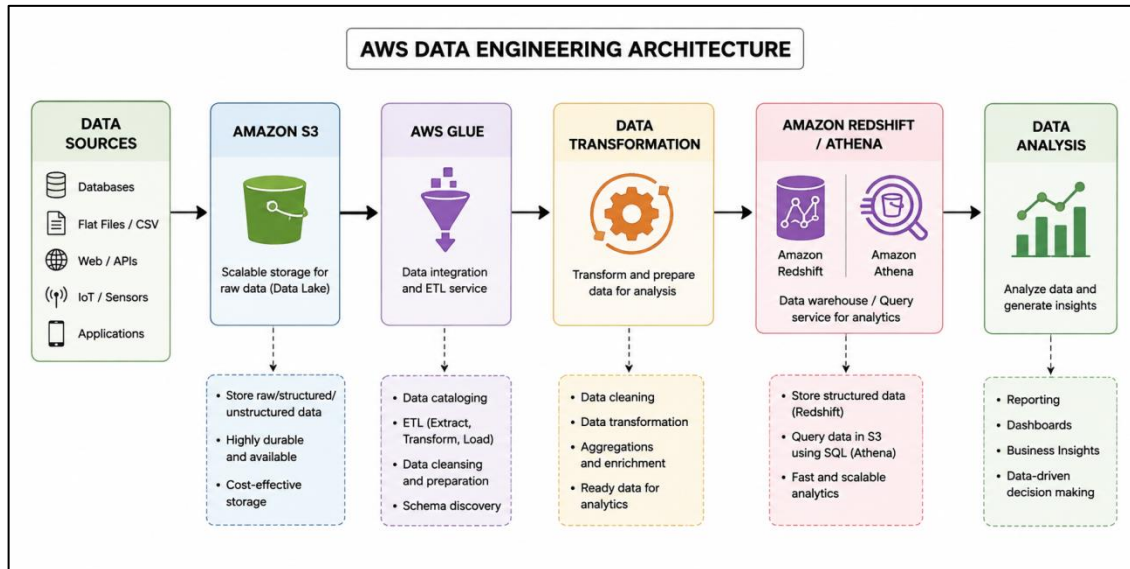


Fig 3 – AWS Data Engineering Architecture

- **Data Sources:** Data may originate from applications, databases, files, websites, or other systems.
- **Amazon S3:** Acts as cloud storage where raw and processed data can be stored.
- **AWS Glue:** Helps in data discovery, integration, ETL, and transformation of data.
- **Amazon Redshift:** Provides a data warehouse environment for storing and analyzing structured data.
- **Amazon Athena:** Allows data stored in Amazon S3 to be queried and analyzed using SQL.

This architecture demonstrates how AWS services can be combined to create a scalable and efficient data engineering workflow.

2.8 Methodology

The internship followed a **structured course-based learning methodology**. Since the internship was conducted virtually, the learning process mainly consisted of online course material, practical learning activities, assessments, and revision.

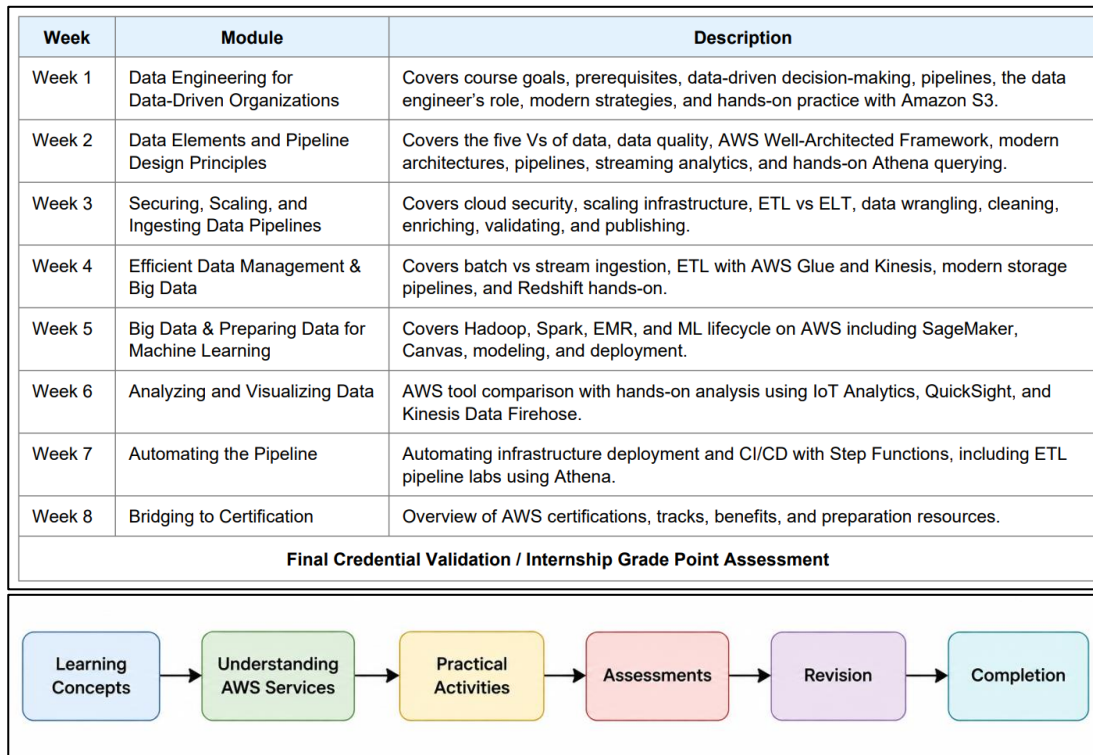


Fig 4 – methodology followed during the internship can be summarized as shown in figure

Initially, the fundamental concepts of cloud computing and AWS were studied. This was followed by learning about data engineering concepts such as data storage, ETL, data processing, and data pipelines.

Practical activities and examples were used to improve understanding of the concepts. Assessments and quizzes were completed to evaluate the knowledge gained during the training.

The internship was completed by reviewing the course material and fulfilling the required training and assessment activities. This approach provided both theoretical knowledge and practical exposure to AWS Data Engineering concepts.

2.9 Expected Outcomes

The AWS Data Engineering internship was expected to provide a strong foundation in cloud-based data engineering and AWS technologies. The major expected outcomes of the internship include:

1. Understanding the fundamentals of **AWS and cloud computing**.
2. Developing knowledge of **Data Engineering concepts and workflows**.
3. Understanding cloud-based data storage and processing.
4. Gaining knowledge of **ETL and data transformation**.
5. Understanding the role of different AWS services in data engineering.
6. Developing basic understanding of **data pipelines and data integration**.
7. Understanding how data can be prepared for analytics and decision-making.

8. Gaining industry-oriented technical knowledge and practical exposure.
9. Improving familiarity with cloud technologies and data-related tools.
10. Building a foundation for further learning and development in **AWS, Cloud Computing, and Data Engineering.**

Overall, the internship was expected to improve technical knowledge and provide a better understanding of how modern organizations use cloud technologies to manage and process data.

3. Internship Supporting Documents

3.1 Internship Offer Letter

This section contains the official offer letter received for the AWS Data Engineering Virtual Internship through EduSkills Foundation. The offer letter confirms the selection and provides the relevant details regarding the internship program.

EduSkills®
Nation Building Through Skills

Ref No: C17Y2026M081450606

Internship Offer Letter

Student Name : Namita Sharma
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering (Cyber Security))
Institute Name : IILM University, Greater Noida
Internship Domain : AWS Data Engineering
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU6982cd6be97bf1770179947

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

Fig 5- Internship Offer Letter

3.2 Internship Completion Confirmation Email

The following email was received from EduSkills Foundation confirming the successful completion of the **10-week AWS Data Engineering Internship**. It also provided information regarding the availability and download of the internship completion certificate. The email is attached as supporting proof of internship completion.

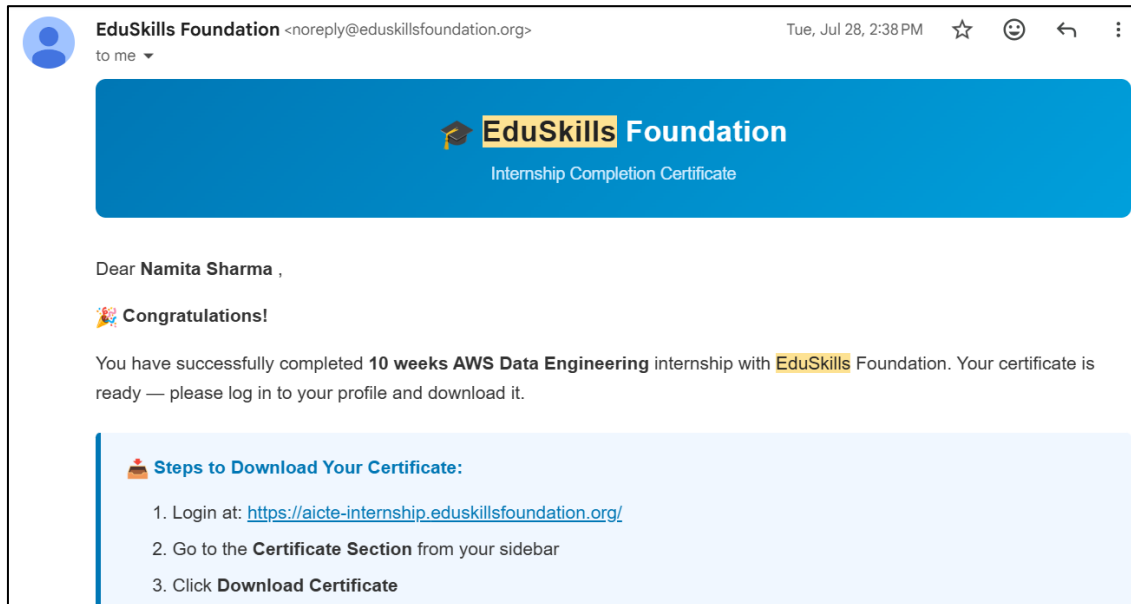


Fig 6 – Internship Completion Confirmation Email

4. References

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